

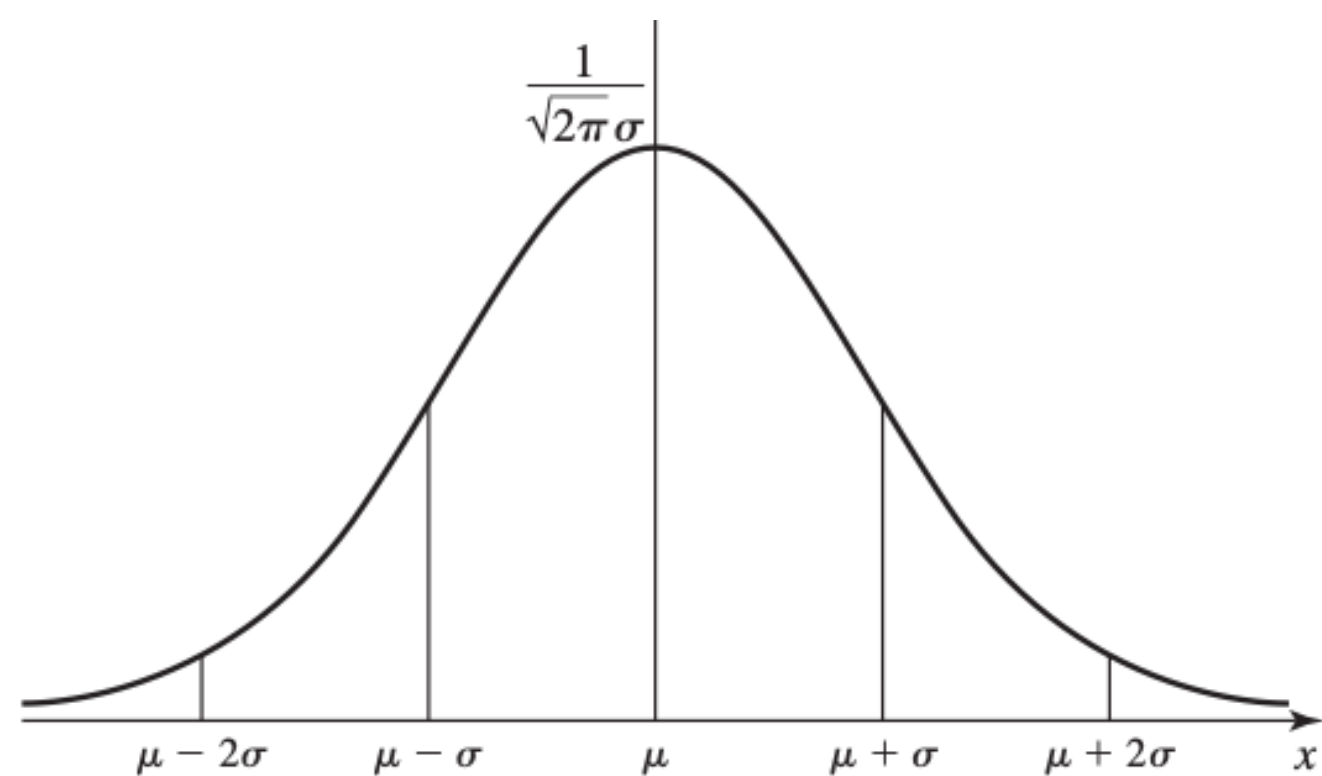
QUIZ

True or False:

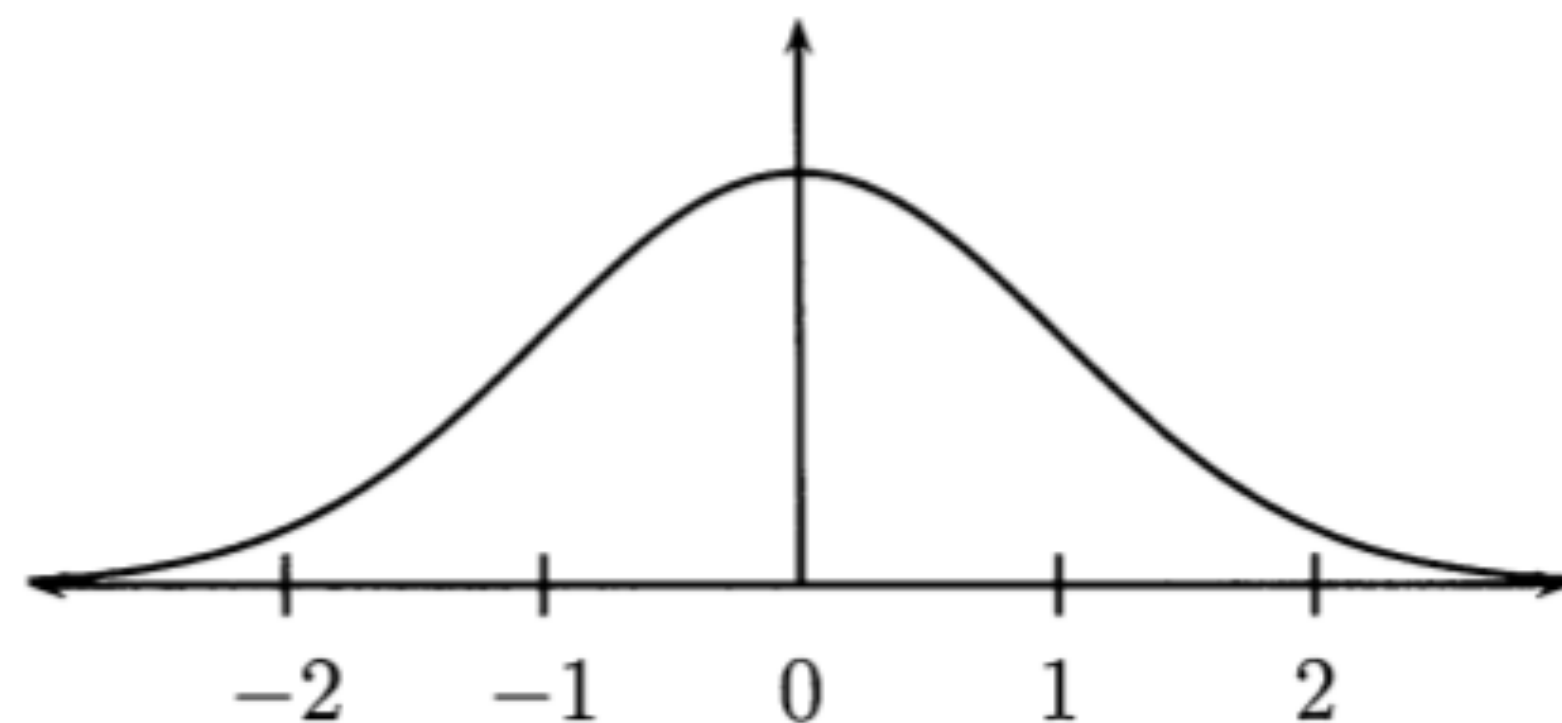
If two random variables are independent, then they must be uncorrelated.

Explain with one sentence.

REVIEW OF (LAST PART OF) THE PREVIOUS LECTURE



$$X \sim N(\mu, \sigma^2)$$

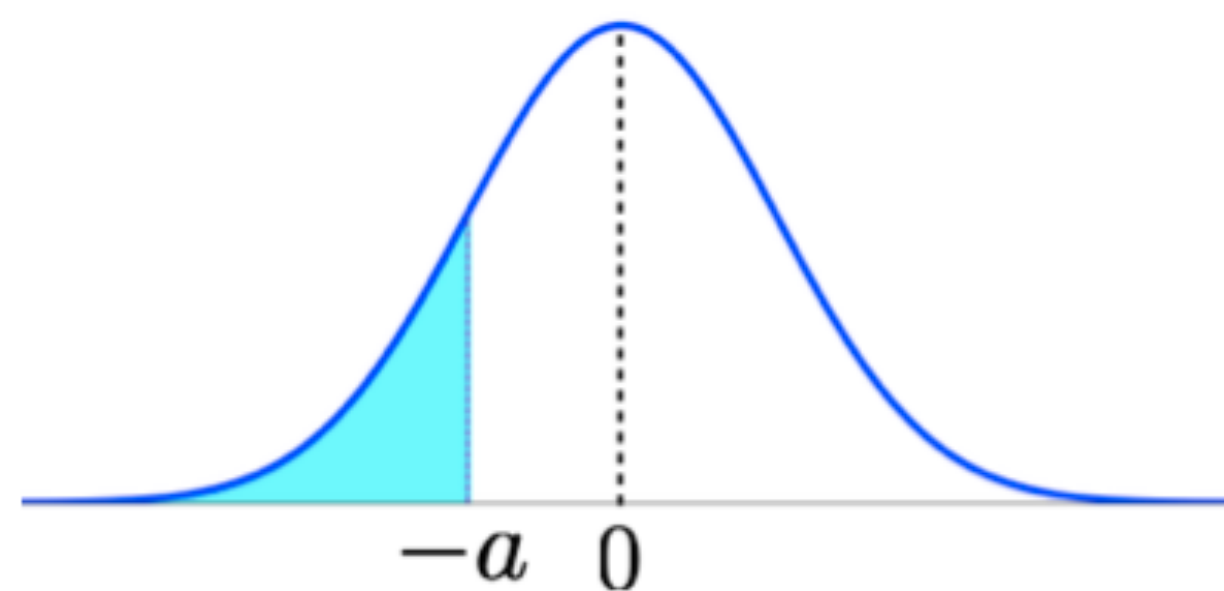


$$Z \sim N(0, 1)$$

Normal to standard normal:

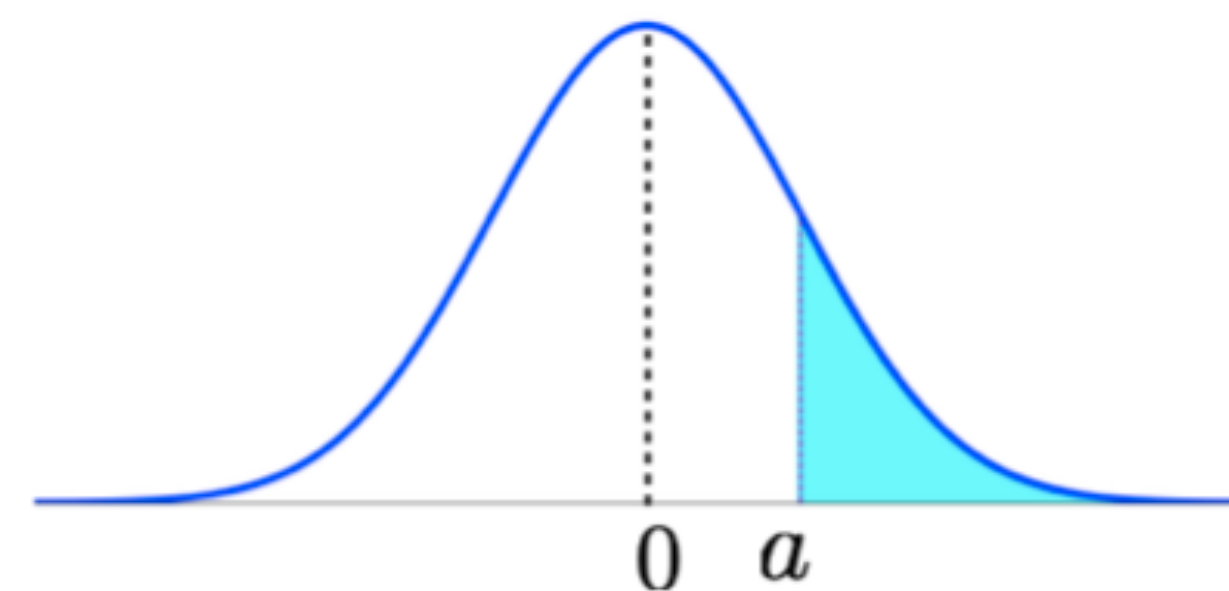
$$X \sim N(\mu, \sigma^2) \Rightarrow$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$



$$\Phi(-a)$$

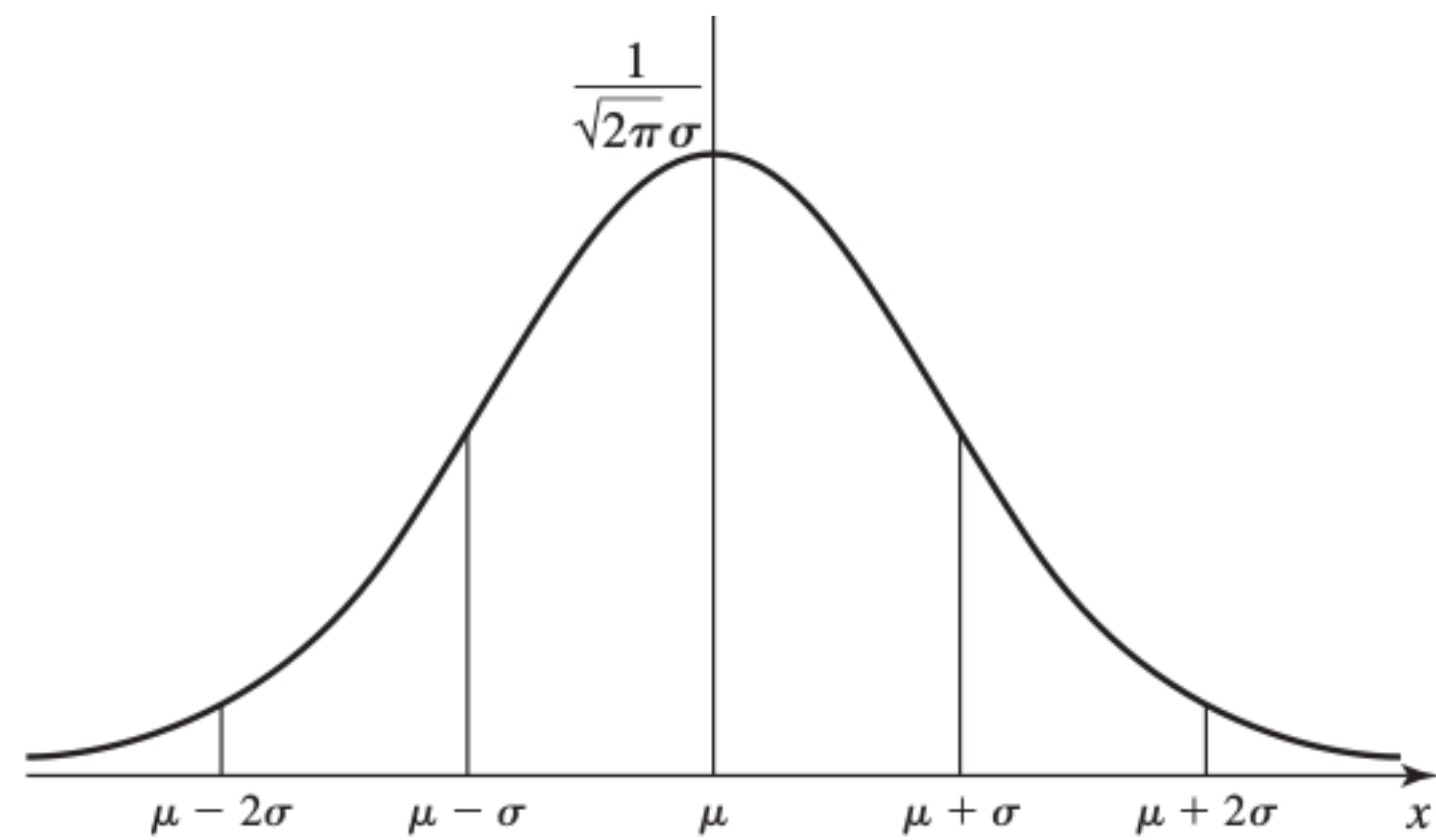
=



$$1 - \Phi(a)$$

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

Easy to find: Probability that normal X within k std of mean.



k	p_k
1	0.6826
2	0.9544
3	0.9974
4	0.99994

p_k : Probability that normal RV within $k \times \sigma$ of μ

How? $P(\mu - k\sigma \leq X \leq \mu + k\sigma) = P(-k \leq Z \leq k)$

$$= 2\Phi(k) - 1$$

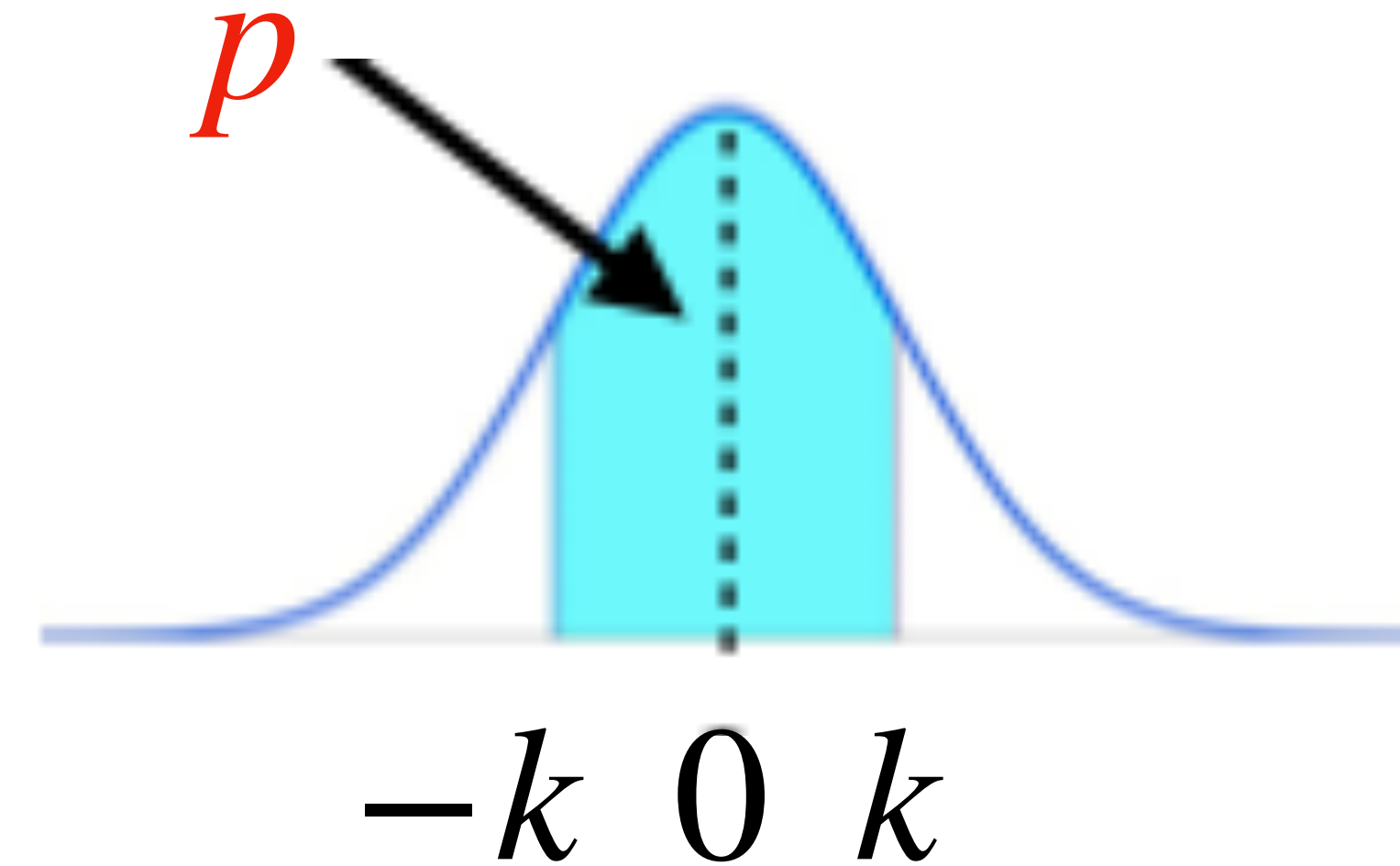
THE QUANTILE FUNCTION

Common problem:

Given desired probability p , find k s.t. $P(-k \leq Z \leq k) = p$

$$p = P(-k \leq Z \leq k) = 2\Phi(k) - 1$$

$$\Rightarrow \Phi(k) = \frac{p + 1}{2} \text{ and } \Phi^{-1}\left(\frac{p + 1}{2}\right) = k$$



THE QUANTILE FUNCTION

```
from scipy.stats import norm  
norm.ppf(0.9)
```

1.2815515655446004

$$\Phi^{-1}(0.9) \approx 1.28 \quad \Rightarrow \quad \Phi(1.28) = P(Z \leq 1.28) \approx 0.9$$

$$\Phi^{-1}(p) : \text{norm.ppf}(p)$$

Percent point function

Can use the *ppf* of other distributions to compute their $F^{-1}(p)$.

THE QUANTILE FUNCTION

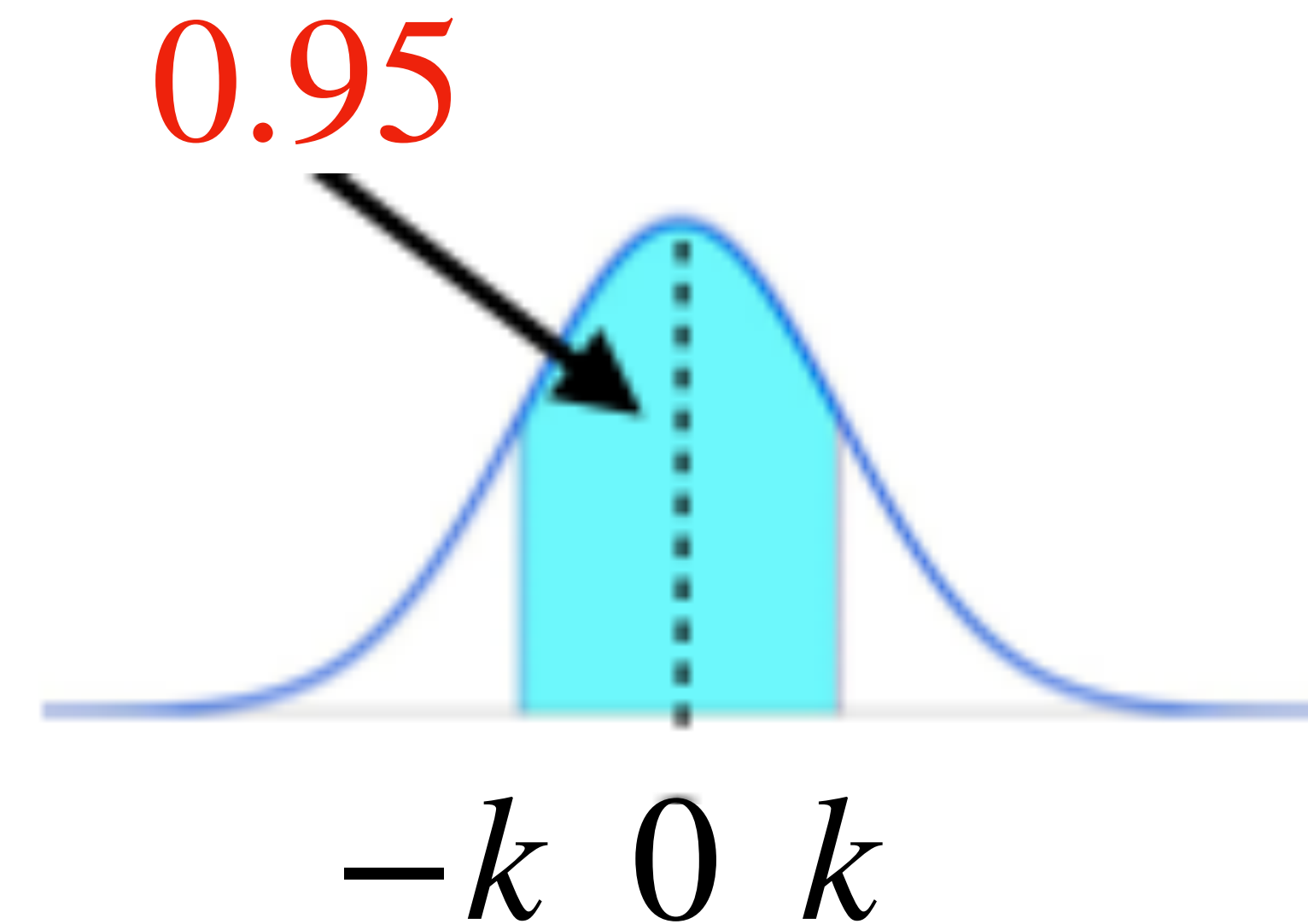
Example: Find k s.t. $P(-k \leq Z \leq k) = 0.95$

$$p = 0.95$$

$$k = \Phi^{-1}\left(\frac{p+1}{2}\right) = \Phi^{-1}(0.975)$$

```
from scipy.stats import norm  
norm.ppf(0.975)
```

1.959963984540054



THE QUANTILE FUNCTION

How to generate synthetic data from a distribution with cdf F ?

Inverse transform:

Repeatedly pick random p in $[0..1]$ and return $F^{-1}(p)$.

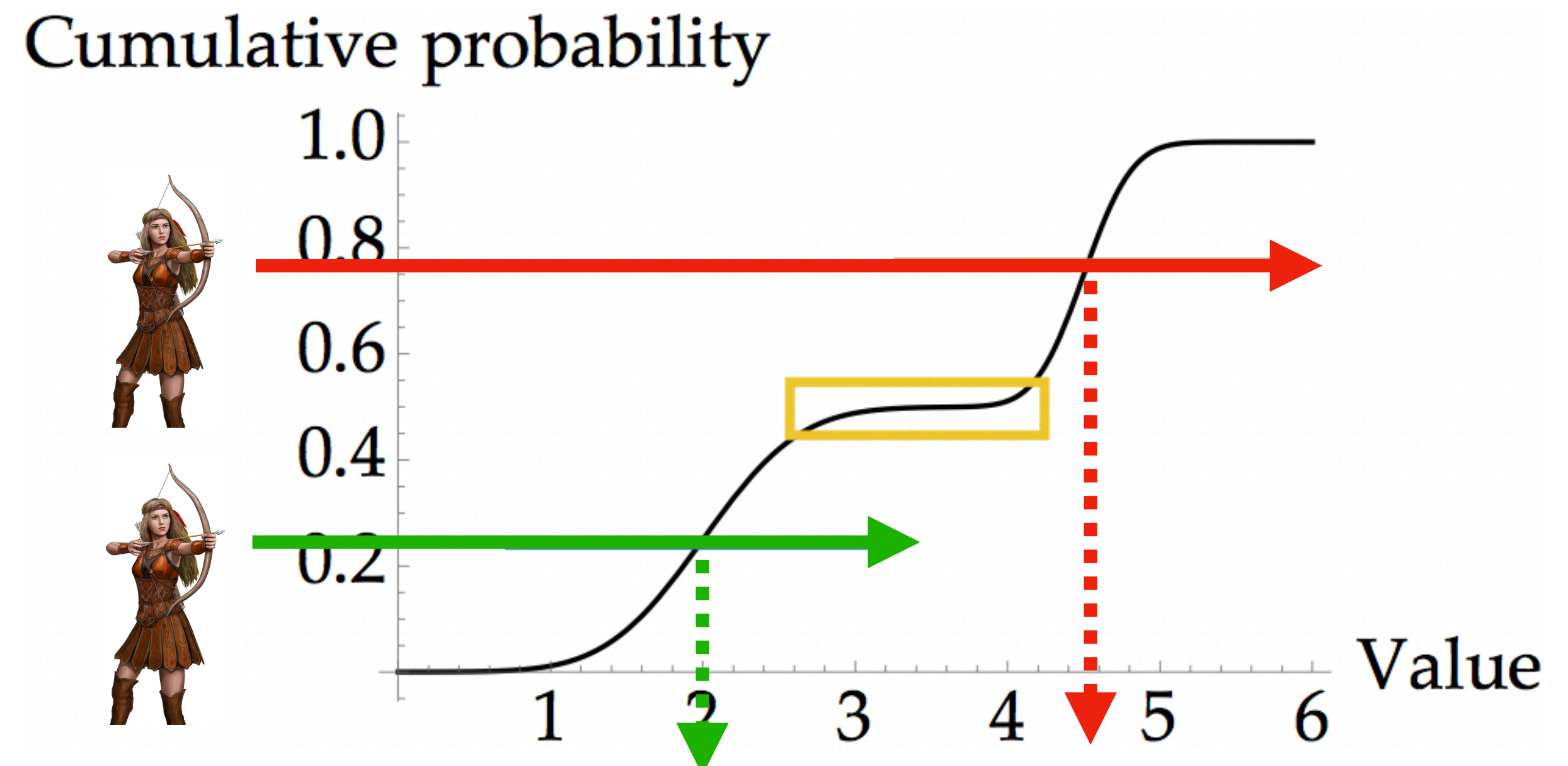
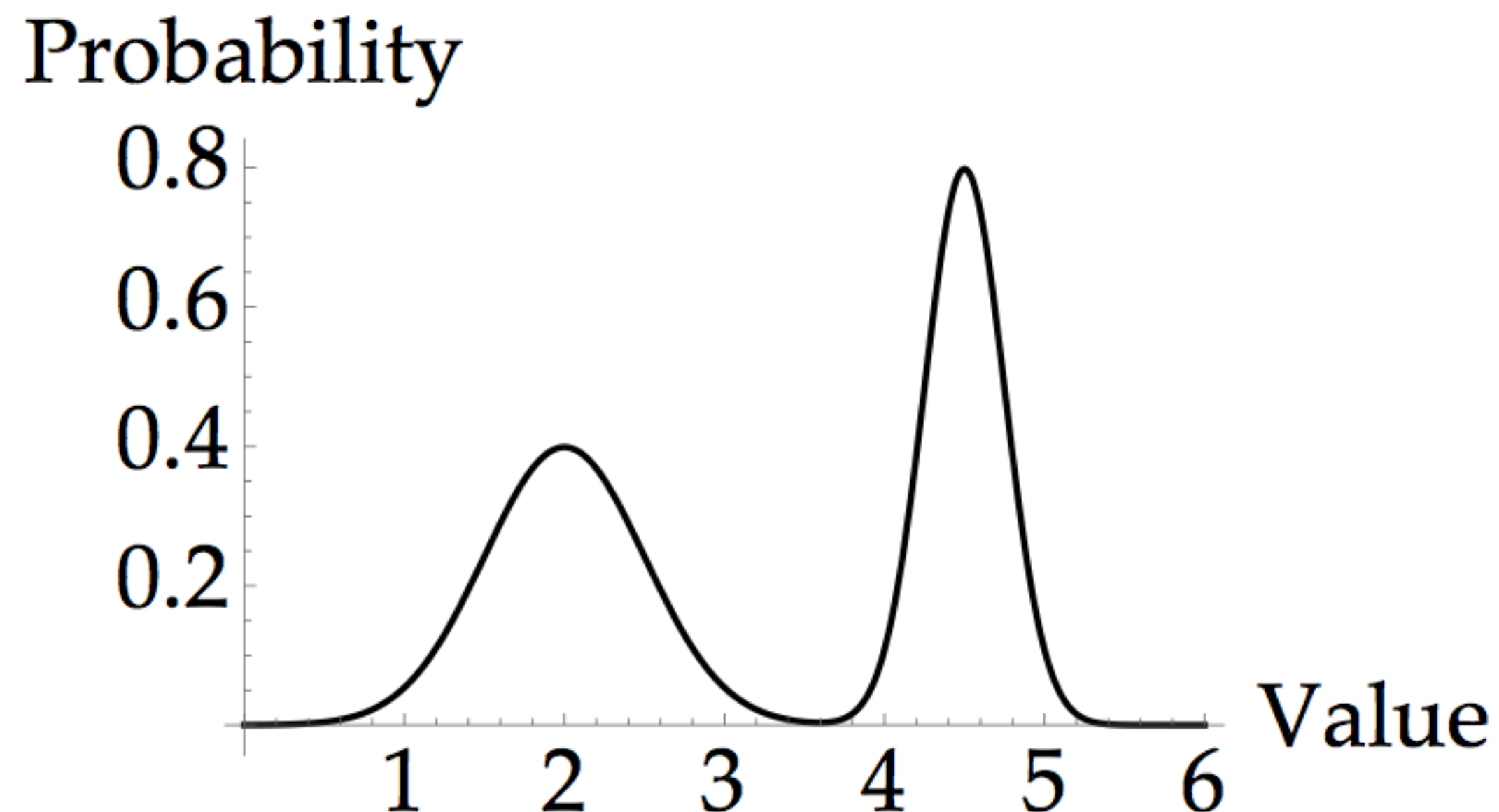
THE QUANTILE FUNCTION

How to generate synthetic data from a distribution with cdf F ?

Inverse transform:

Repeatedly pick random p in $[0..1]$ and return $F^{-1}(p)$.

Why does it work?



PROBABILITY INEQUALITIES

Markov's Inequality:

X a non-negative random variable $\Rightarrow \forall t > 0, P(X \geq t) \leq \frac{\mu}{t}$

Proof:

X discrete random variable. Similar proof works for continuous.

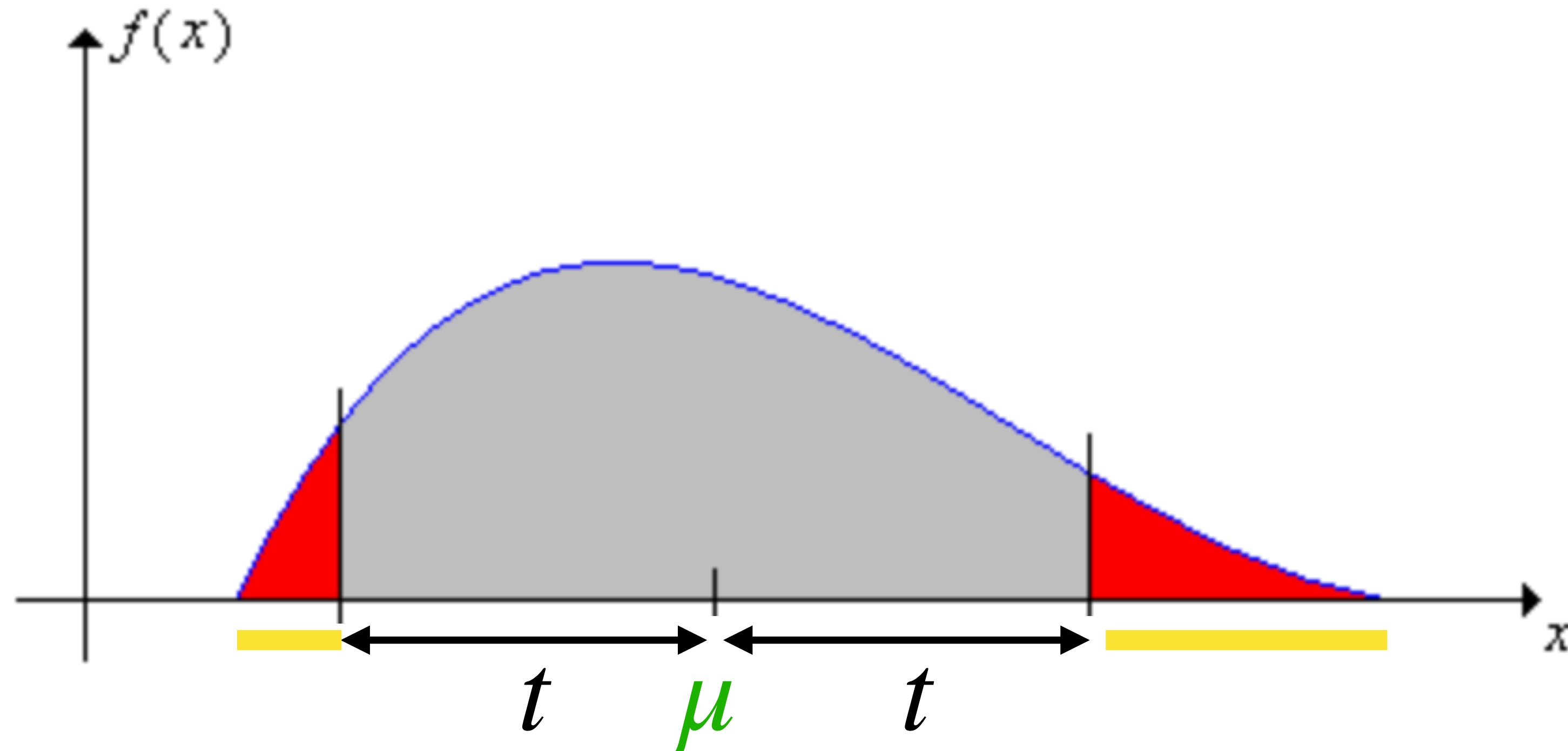
$$\mu = \sum_{\forall x} xf(x) \geq \sum_{x \geq t} xf(x) \geq \sum_{x \geq t} tf(x) = t \sum_{x \geq t} f(x) = tP(X \geq t)$$

Interesting only for $t > \mu$.

PROBABILITY INEQUALITIES

Chebyshev's Inequality:

X a random variable $\Rightarrow \forall t > 0, P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}$

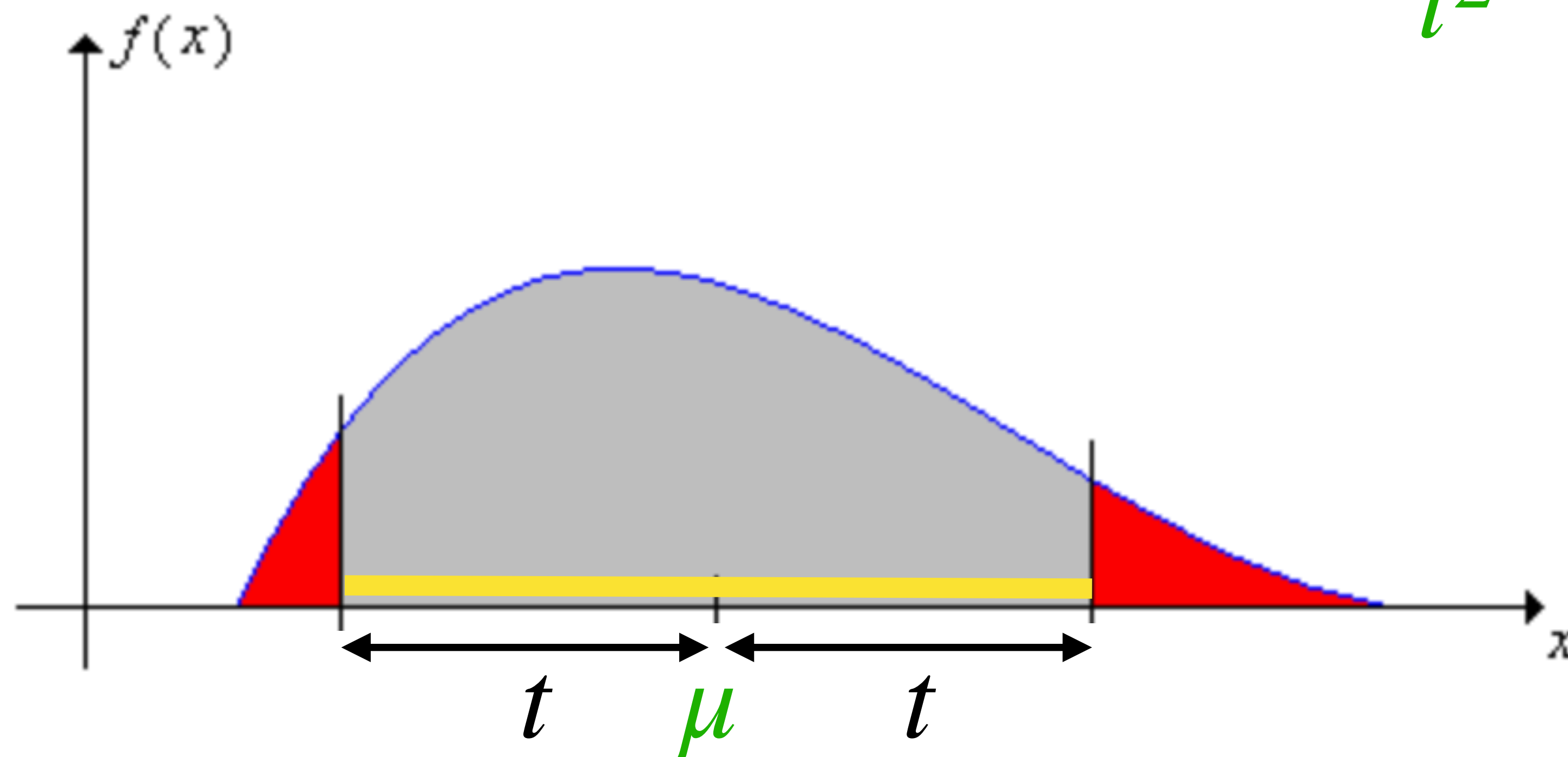


$P(X \text{ lies in yellow interval}) \leq \frac{\sigma^2}{t^2}$ (area in red)

PROBABILITY INEQUALITIES

Chebyshev's Inequality: Another interpretation

$$P(|X - \mu| < t) \geq 1 - \frac{\sigma^2}{t^2}$$

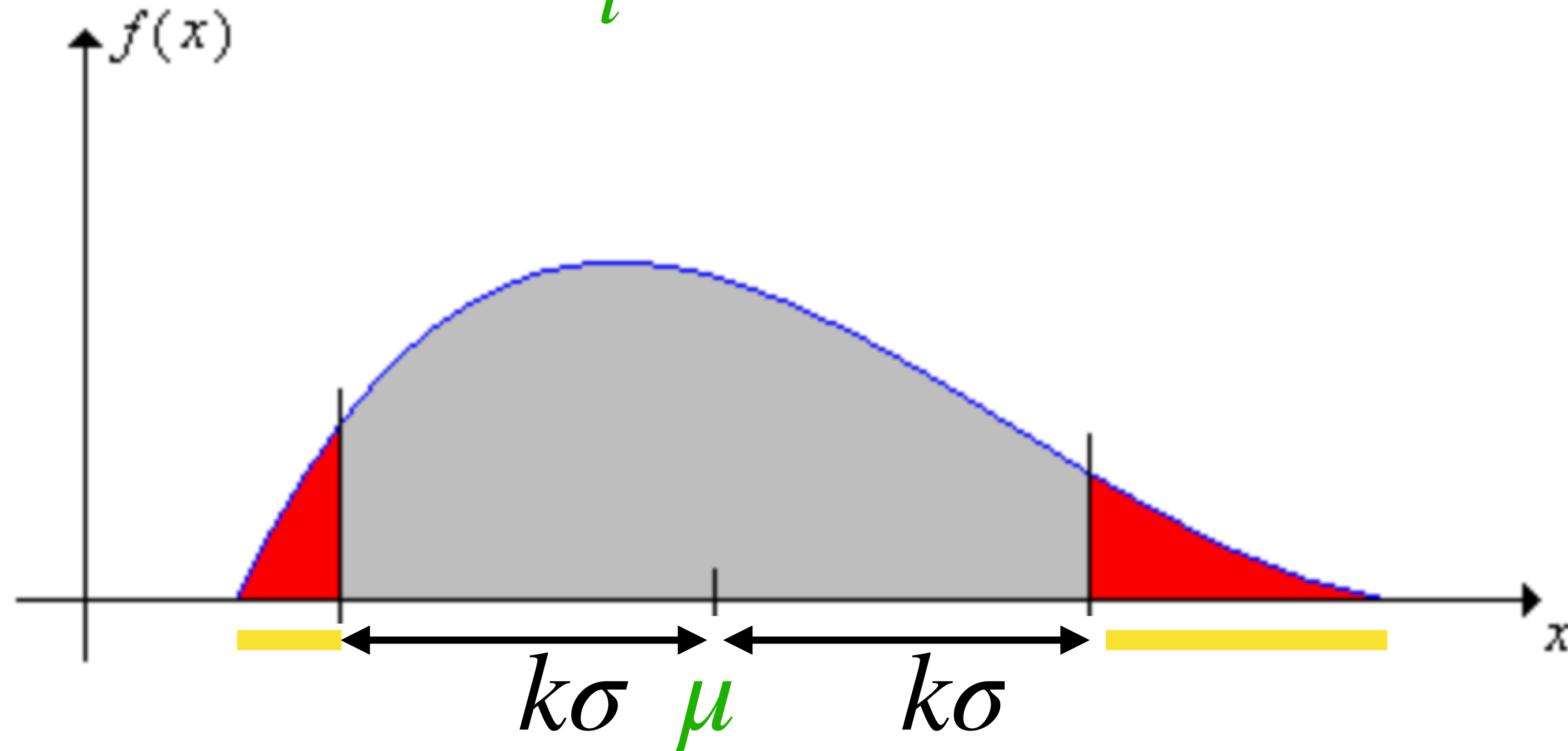


$$P(X \text{ lies in yellow interval}) \geq 1 - \frac{\sigma^2}{t^2} \quad (\text{area in gray})$$

PROBABILITY INEQUALITIES

Chebyshev's Inequality: Being at least k std away from mean

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2} \quad \Rightarrow \quad P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

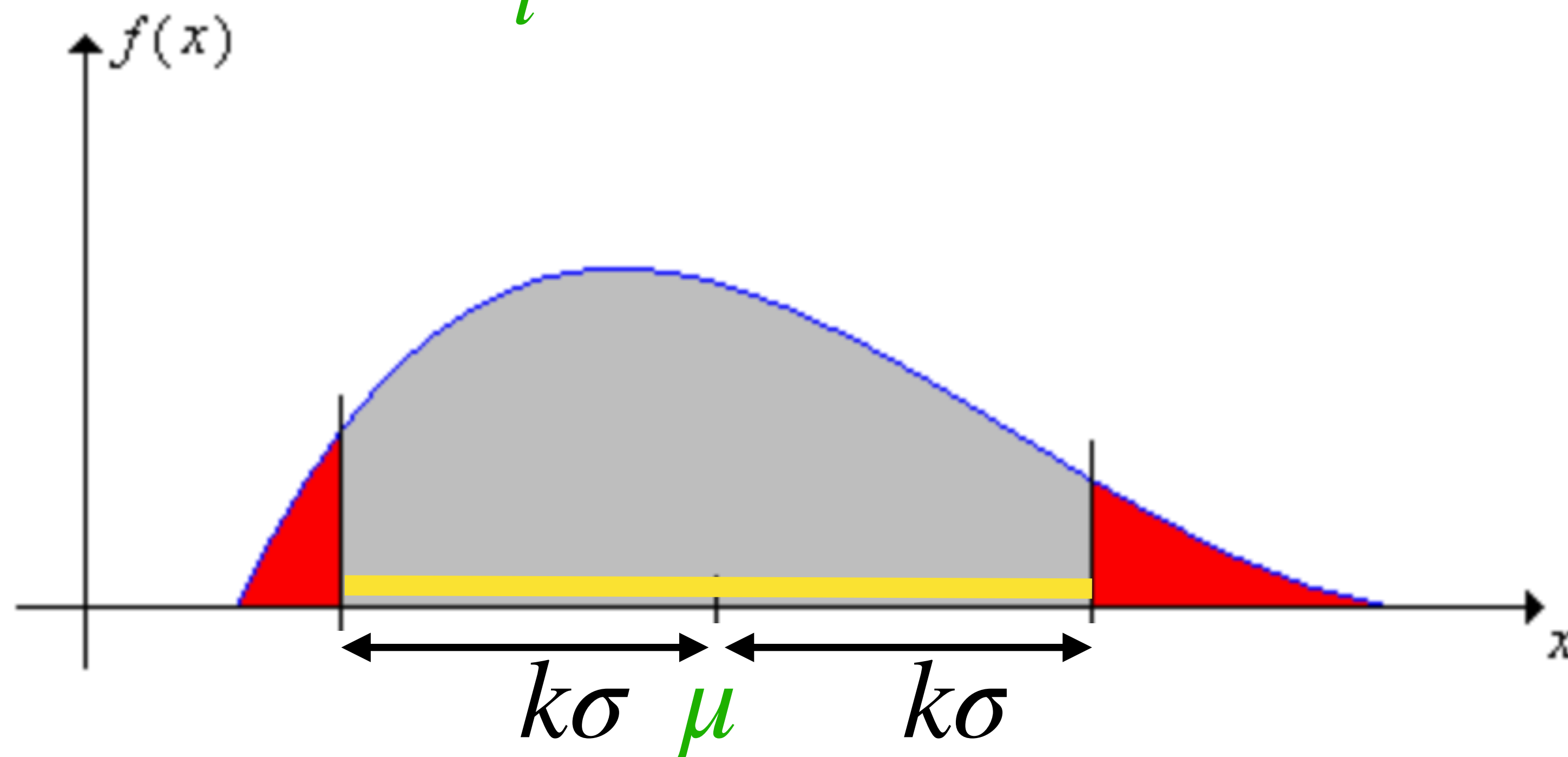


$$P(X \text{ lies in yellow interval}) \leq \frac{1}{k^2} \quad (\text{area in red})$$

PROBABILITY INEQUALITIES

Chebyshev's Inequality: Being within k std of the mean

$$P(|X - \mu| < t) \geq 1 - \frac{\sigma^2}{t^2} \quad \Rightarrow \quad P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$



$$P(X \text{ lies in yellow interval}) \geq 1 - \frac{1}{k^2} \quad (\text{area in gray})$$

PROBABILITY INEQUALITIES

Chebyshev's Inequality:

X a random variable $\Rightarrow \forall t > 0, P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}$

Proof:

$$P(|X - \mu| \geq t) = P((X - \mu)^2 \geq t^2) \leq \frac{E[(X - \mu)^2]}{t^2} = \frac{\sigma^2}{t^2}$$

PROBABILITY INEQUALITIES

Example: Let X denote the number of citations of a paper.

Let $\mu = 8$, $\sigma = 5$. What is $P(X \geq 58)$?

With Markov's Inequality:

$$P(X \geq 58) \leq \frac{8}{58} \approx 0.14$$

With Chebyshev's Inequality:

$$P(X \geq 58) = P(X - \mu \geq 50) \leq P(|X - \mu| \geq 50) \leq \left(\frac{\sigma}{50}\right)^2 = 0.01$$

SAMPLE MEAN

$X_1, \dots, X_n$ iid with mean μ and variance σ^2 . Sample mean $\bar{X}_n$

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Note that $\bar{X}_n$ is a random variable.

$$E(\bar{X}_n) = E\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n} \sum E(X_i) = \frac{\mu n}{n} = \mu$$

$$Var(\bar{X}_n) = Var\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n^2} \sum Var(X_i) = \frac{\sigma^2 n}{n^2} = \frac{\sigma^2}{n}$$

SAMPLE MEAN

Example: Poll 100,000 people. Assume each votes for candidate C independently with probability p .

Bound the probability that the poll is off by more than 1 % .

$X_i = 1$ if i votes for C , $X_i = 0$ o.w. $X_i \sim \text{Bernoulli}(p)$

First find a bound for σ^2 . What is max possible value for it?

$\mu = p$, $\sigma^2 \leq 1/4$ since $p(1 - p)$ is maximum when $p = 1/2$

[find p where derivative of $p(1 - p)$ is 0]